

Vehicle Showroom Management System

Complete Unabridged 18-Class UML Class Specification Document

1. Unabridged Class Specifications (All Attributes & Methods Preserved)

<<Abstract>>

Vehicle

```
# modelID: string
# brand: string
# modelName: string
# year: int
# color: string
# price: double
# quantity: int
# isElectric: bool
# engineCC: int
# mileageKmpl: double
# fuelType: string

+ Vehicle()
+ Vehicle(modelID: string, brand: string, modelName: string, year: int, color: string, price: double, quantity: int, isElectric: bool, engineCC: int, mileageKmpl: double, fuelType: string)
+ virtual ~Vehicle()
+ virtual displayInfo() const = 0: void
+ virtual getType() = 0: string
+ getModelID() const: string
+ getBrand() const: string
+ getModelName() const: string
+ getPrice() const: double
+ getQuantity() const: int
+ setQuantity(q: int): void
```

Car

```
- seatingCapacity: int
- transmission: string
- numDoors: int

+ Car()
+ Car(modelID: string, brand: string, modelName: string, year: int, color: string, price: double, quantity: int, isElectric: bool, engineCC: int, mileageKmpl: double, fuelType: string, _seatingCapacity: int, _transmission: string, _numDoors: int)
+ Car(const Car& other)
+ displayInfo() const override: void
+ getType() override: string
+ getSeatingCapacity() const: int
+ getTransmission() const: string
+ getNumDoors() const: int
+ static getCarModel(__modelID: const string&): Car*
```

Bike

```
- bikeType: string

+ Bike()
+ Bike(modelID: string, brand: string, modelName: string, year: int, color: string, price: double, quantity: int, isElectric: bool, engineCC: int, mileageKmpl: double, fuelType: string, _bike_type: string)
+ Bike(const Bike& other)
+ displayInfo() const override: void
+ getType() override: string
+ get_bikeType() const: string
+ static getBikeModel(__modelID: const string&): Bike*
```

Scooty

```
- scootyType: string

+ Scooty()
+ Scooty(modelID: string, brand: string, modelName: string, year: int, color: string, price: double,
quantity: int, isElectric: bool, engineCC: int, mileageKmpl: double, fuelType: string, _scootyType: string)
+ Scooty(const Scooty& other)
+ displayInfo() const override: void
+ getType() override: string
+ get_scootyType() const: string
+ static getScootyModel(__modelID: const string&): Scooty*
```

SoldVehicle

```
- plateNumber: string
- purchaseDate: string
- baseVehicle: Vehicle*

+ SoldVehicle()
+ SoldVehicle(_plate: string, _date: string, _v: Vehicle*)
+ getPlateNumber() const: string
+ getPurchaseDate() const: string
+ getVehicle() const: Vehicle*
```

<<Abstract>>

Person

```
# id: string
# name: string
# email: string
# phone: string
# username: string
# password: string

+ Person()
+ Person(_id: string, _name: string, _email: string, _phone: string, _username: string, _password: string)
+ Person(const Person& other)
+ virtual ~Person()
+ getID() const: string
+ getName() const: string
+ getEmail() const: string
+ getPhone() const: string
+ getUsername() const: string
+ checkPassword(inp_pass: const string&) const: bool
+ virtual showMenu() = 0: void
+ virtual getRole() const = 0: string
+ virtual disp_Info() const = 0: void
```

Employee

```
# cnic: string
# hireDate: string
# salary: double

+ Employee()
+ Employee(_id: string, _name: string, _email: string, _phone: string, _username: string, _password: string,
_cnic: string, _hireDate: string, _salary: double)
+ Employee(const Employee& other)
+ getCNIC() const: string
+ getHireDate() const: string
+ getSalary() const: double
+ setSalary(nw_sal: double): void
```

RepairMan

```
- specialization: string

+ RepairMan()
+ RepairMan(_id: string, _name: string, _email: string, _phone: string, _username: string, _password: string,
_cnic: string, _hireDate: string, _Salary: double, _specialization: string)
+ RepairMan(const RepairMan& other)
+ showMenu() override: void
+ getRole() const override: string
+ getSpecialization() const: string
+ getPasscode() const: string
+ disp_Info() const override: void
+ viewPendingJobs(rm: RepairManager&) const: void
+ acceptJob(rm: RepairManager&, repairID: const string&): void
+ markJobCompleted(rm: RepairManager&, repairID: const string&): void
+ static get_RepairmanById(__id: string): RepairMan*
```

Salesman

```
- totalSalesCount: int
- commissionRate: double

+ Salesman()
+ Salesman(_id: string, _name: string, _email: string, _phone: string, _username: string, _password: string,
_cnic: string, _hireDate: string, _Salary: double, _salesCount: int, _comm: double)
+ Salesman(const Salesman& other)
+ showMenu() override: void
+ getRole() const override: string
+ disp_Info() const override: void
+ recordSale(sm: SalesManager&, inv: Inventory&, custID: string, modelID: string, finalPrice: double): void
+ getSalesCount() const: int
+ getCommissionRate() const: double
+ static get_SalesmanById(__id: string): Salesman*
```

Customer

```
- cnic: string
- address: string
- registration_date: string

+ Customer()
+ Customer(_id: string, _name: string, _email: string, _phone: string, _username: string, _password: string,
_cnic: string, _address: string, _registration_date: string)
+ Customer(const Customer& other)
+ showMenu() override: void
+ getRole() const override: string
+ getPasscode() const: string
+ get_Date() const: string
+ disp_Info() const override: void
+ getCNIC() const: string
+ getAddress() const: string
+ browse_cars(inv: Inventory&) const: void
+ browse_bikes(inv: Inventory&) const: void
+ browse_scooties(inv: Inventory&) const: void
+ browse_vehicles(inv: Inventory&) const: void
+ viewMyPurchases(sm: SalesManager&) const: void
+ viewMyRepairHistory(rm: RepairManager&) const: void
+ requestVehicleRepair(rm: RepairManager&, plateNumber: const string&, vehicleType: const string&,
partChanged: const string&): void
+ static getCustomer_ById(__id: string): Customer*
```

Admin

- id: string
- name: string
- username: string
- password: string

```
+ Admin()
+ Admin(_id: string, _name: string, _username: string, _password: string)
+ showMenu(): void
+ getRole() const: string
+ getUsername() const: string
+ checkPassword(inp: const string&) const: bool
+ addCar(inv: Inventory&): void
+ addBike(inv: Inventory&): void
+ addScooty(inv: Inventory&): void
+ removeVehicleModel(inv: Inventory&, modelID: const string&): void
+ updateStock(inv: Inventory&, modelID: const string&, newQuantity: int): void
+ addSalesman(salesmen: vector<Salesman>&): void
+ removeSalesman(salesmen: vector<Salesman>&, salesmanID: const string&): void
+ addRepairMan(repairmen: vector<RepairMan>&): void
+ removeRepairMan(repairmen: vector<RepairMan>&, repairManID: const string&): void
+ paySalary(emp: Employee&, amount: double): void
+ addCustomer(customers: vector<Customer>&): void
+ removeCustomer(customers: vector<Customer>&, customerID: const string&): void
+ viewSalesReport(sm: SalesManager&) const: void
+ viewAllVehicles(inv: Inventory&) const: void
+ viewAllSalesmen(salesmen: const vector<Salesman>&) const: void
+ viewAllRepairmen(repairmen: const vector<RepairMan>&) const: void
+ viewAllCustomers(customers: const vector<Customer>&) const: void
+ viewPendingRepairs(rm: RepairManager&) const: void
+ static getAdmin(): Admin*
```

Inventory

- cars: vector<Car>
- bikes: vector<Bike>
- scooties: vector<Scooty>
- intro_lines: vector<string>

```
+ loadAll(): void
+ Changeable_disp(type: string = "NULL"): void
+ saveAll() const: void
+ getAllVehicles(): vector<Vehicle*>
+ findByModelID(modelID: const string&): Vehicle*
+ addCar(c: const Car&): void
+ addBike(b: const Bike&): void
+ addScooty(s: const Scooty&): void
+ removeModel(modelID: const string&): bool
+ updateQuantity(modelID: const string&, newQuantity: int): void
+ getCars(): vector<Car*>
+ getBikes(): vector<Bike*>
+ getScooties(): vector<Scooty*>
```

Part

- name_of_part: string
- cost_of_part: double

```
+ Part()
+ Part(_name: string, _cost: double)
+ get_Part() const: string
+ get_Cost() const: double
+ change_Part_Name(nw_name: string): void
+ change_Cost(nw_cost: double): void
```

RepairRecord

- repairID: string
- customerID: string
- plateNumber: string
- vehicleType: string
- partChanged: string
- status: string
- assignedRepairManID: string
- date: string
- partCost: double
- laborCost: double
- totalCost: double

```
+ RepairRecord()
+ RepairRecord(_repID: string, _custID: string, _plate: string, _vType: string, _part: string, _status:
string, _repManID: string, _date: string, _pCost: double, _lCost: double)
+ getRepairID() const: string
+ getCustomerID() const: string
+ getPlateNumber() const: string
+ getVehicleType() const: string
+ getPartChanged() const: string
+ getStatus() const: string
+ getAssignedRepairManID() const: string
+ getDate() const: string
+ getPartCost() const: double
+ getLaborCost() const: double
+ getTotalCost() const: double
+ setStatus(_status: string): void
+ setRepairMan(_repManID: string): void
```

SaleRecord

- transactionID: string
- customerID: string
- salesmanID: string
- modelID: string
- finalPrice: double
- date: string

```
+ SaleRecord()
+ SaleRecord(_tID: string, _cID: string, _sID: string, _mID: string, _price: double, _date: string)
+ getTransactionID() const: string
+ getCustomerID() const: string
+ getSalesmanID() const: string
+ getModelID() const: string
+ getFinalPrice() const: double
+ getDate() const: string
```

RepairManager

- repairs: vector<RepairRecord>
- partPriceList: vector<Part>
- intro_lines: vector<string>

```
+ loadAll(): void
+ saveAll() const: void
+ getPartCost(part_name: const string&) const: double
+ requestRepair(customerID: const string&, plateNumber: const string&, vehicleType: const string&,
partChanged: const string&, date: string&): string
+ getHistoryByCustomer(customer_ID: const string&) const: vector<RepairRecord>
+ getJobsByRepairMan(Repair_ManID: const string&) const: vector<RepairRecord>
+ getPendingJobs() const: vector<RepairRecord>
+ assignRepairMan(__repair_ID: const string&, repairManID: const string&): bool
+ updateStatus(repair_id: const string&, newSts: const string&): bool
```

SalesManager
- salesHistory: vector<SaleRecord> - soldVehiclesRegistry: vector<SoldVehicle> - intro_lines: vector<string>
+ loadAll(): void + saveAll() const: void + addSaleRecord(record: SaleRecord): void + addSoldVehicle(sv: SoldVehicle): void + getSalesHistory() const: vector<SaleRecord> + getSoldVehicles() const: vector<SoldVehicle> + verifyOwnership(custID: string, plate: string) const: bool

ShowroomSystem
- inventory: Inventory - salesMgr: SalesManager - repairMgr: RepairManager - systemLogs: vector<string>
+ ShowroomSystem() + initializeSystem(): void + runMainWorkflow(): void + authenticateUser(): void + shutdownGracefully(): void

2. Structural System Relationships Table

Source Class	Relationship Type	Target Class	Description
Car, Bike, Scooty	Inheritance (⇐)	Vehicle	Generalization of standard showroom assets
Employee, Customer	Inheritance (⇐)	Person	Generalization of human components
RepairMan, Salesman	Inheritance (⇐)	Employee	Refined internal working personnel tiers
SoldVehicle	Aggregation (o—)	Vehicle	Dynamic link via polymorphic instance reference
Inventory	Composition (•—)	Car, Bike, Scooty	Main lifetime vehicle model groups
RepairManager	Composition (•—)	RepairRecord, Part	Main maintenance management lists
SalesManager	Composition (•—)	SaleRecord, SoldVehicle	Main commercial ledger trackers
ShowroomSystem	Composition (•—)	Inventory, SalesManager, RepairManager	Core central workflow orchestration system